

$$1. -2A = \begin{bmatrix} -4 & 0 & 2 \\ -8 & -3 & -4 \end{bmatrix} \quad AC = \begin{bmatrix} 2 & 0 & -1 \\ 4 & -3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \quad \text{not same size}$$

$\Rightarrow \text{undefined}$

$$B - 2A = \begin{bmatrix} 7 & -5 & 1 \\ 1 & -4 & -3 \end{bmatrix} + \begin{bmatrix} -4 & 0 & 2 \\ -8 & -3 & -4 \end{bmatrix} = \begin{bmatrix} 3 & -5 & 3 \\ -7 & -7 & -7 \end{bmatrix}$$

Linear Algebra

線性代數

Homework 3

$$CD = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 13 \\ -7 & -6 \end{bmatrix}$$

1. Compute each matrix sum or product if it is defined. If an expression is undefined, explain why.

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & -3 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & -5 & 1 \\ 1 & -4 & -3 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 5 \\ -1 & 4 \end{bmatrix}, \quad E = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$$

1. $-2A$, $B - 2A$, AC , CD

2. Assume that each matrix expression is defined. That is, the sizes of the matrices (and vectors) involved "match" appropriately.

Compute $A - 5I_3$ and $(5I_3)A$, when

$$A = \begin{bmatrix} 9 & -1 & 3 \\ -8 & 7 & -3 \\ -4 & 1 & 8 \end{bmatrix} \quad A - 5I_3 = \begin{bmatrix} 4 & -1 & 3 \\ -8 & 2 & -3 \\ -4 & 1 & 3 \end{bmatrix}$$

3. Compute the product AB in two ways: (a) by the definition, where Ab_1 and Ab_2 are computed separately, and (b) by the row-column rule for computing AB .

$$A = \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -4 \\ -2 & 1 \end{bmatrix}$$

4. If a matrix A is 5×3 and the product AB is 5×7 , what is the size of B ?

$$A_{5 \times 3} B_{?} = AB_{5 \times 7}$$

$\uparrow$
 $m \times n$

$$m=3$$

$$n=7$$

#

$$= \begin{bmatrix} 45 & -5 & 15 \\ -40 & -35 & -15 \\ -20 & 5 & 40 \end{bmatrix} \#$$

5.

Let $A = \begin{bmatrix} 3 & 4 \\ -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & -6 \\ 3 & k \end{bmatrix}$. What value(s) of k , if any, will make $AB = BA$?

$$AB = \begin{bmatrix} 3 & 4 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 5 & -6 \\ 3 & k \end{bmatrix} = \begin{bmatrix} 22 & -18+4k \\ -7 & 12+k \end{bmatrix}$$

$$BA = \begin{bmatrix} 5 & -6 \\ 3 & k \end{bmatrix} \begin{bmatrix} 3 & 4 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 27 & 14 \\ 9-2k & 12+k \end{bmatrix}$$

$$-18+4k = 14 \Rightarrow 4k = 32, k = 8 \#$$

6.

3.

(a) $AB = [Ab_1 \quad Ab_2]$

$$Ab_1 = \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -7 \\ 7 \\ 12 \end{bmatrix}$$

$$Ab_2 = \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ -16 \\ -11 \end{bmatrix}$$

$$AB = [Ab_1 \quad Ab_2]$$

$$= \begin{bmatrix} -7 & 6 \\ 7 & -16 \\ 12 & -11 \end{bmatrix}$$

(b) $\overset{AB}{=} \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ -2 & 1 \end{bmatrix}$

$$= \begin{bmatrix} (-1) \times 3 + 2 \times (-2) & (-1) \times (-4) + 2 \times 1 \\ 5 \times 3 + 4 \times (-2) & 5 \times (-4) + 4 \times 1 \\ 2 \times 3 + (-3) \times (-2) & 2 \times (-4) + (-3) \times 1 \end{bmatrix}$$

$$= \begin{bmatrix} -7 & 6 \\ 7 & -16 \\ 12 & -11 \end{bmatrix}$$

6.① AD

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 2 + 1 \times 0 + 1 \times 0 & 1 \times 0 + 1 \times 3 + 1 \times 0 \\ 1 \times 2 + 2 \times 0 + 3 \times 0 & 1 \times 0 + 2 \times 3 + 3 \times 0 \\ 1 \times 2 + 4 \times 0 + 5 \times 0 & 1 \times 0 + 4 \times 3 + 5 \times 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \times 0 + 1 \times 0 + 1 \times 5 \\ 1 \times 0 + 2 \times 0 + 3 \times 5 \\ 1 \times 0 + 4 \times 0 + 5 \times 5 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 5 \\ 2 & 6 & 15 \\ 2 & 12 & 25 \end{bmatrix}$$

DA

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2x_1 + 0x_2 + 0x_3 & 2x_1 + 0x_2 + 0x_3 & 2x_1 + 0x_2 + 0x_3 \\ 0x_1 + 3x_2 + 0x_3 & 0x_1 + 3x_2 + 0x_3 & 0x_1 + 3x_2 + 0x_3 \\ 0x_1 + 0x_2 + 5x_3 & 0x_1 + 0x_2 + 5x_3 & 0x_1 + 0x_2 + 5x_3 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 5 & 20 & 25 \end{bmatrix}$$

(2)

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 1 & 4 & 5 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{x(-1)} \xrightarrow{x(-1)} A^T B = \begin{bmatrix} 1 & \frac{1}{2} & -\frac{1}{2} \\ 1 & -2 & 1 \\ -1 & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & -\frac{1}{2} \\ 1 & -2 & 1 \\ -1 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 1 & -2 & 1 \\ -1 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 3 & 4 & -1 & 0 & 1 \end{bmatrix} \xrightarrow{x(-3)}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & -2 & 2 & -3 & 1 \end{bmatrix} \xrightarrow{x(-1)} \xrightarrow{x(-\frac{1}{2})}$$

$$\sim \begin{bmatrix} 1 & 0 & -1 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \xrightarrow{x(-2)} \xrightarrow{x(1)}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

Ans: $\begin{bmatrix} 1 & \frac{1}{2} & -\frac{1}{2} \\ 1 & -2 & 1 \\ -1 & \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \#$

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$. Com-

pute AD and DA . Explain how the columns or rows of A change when A is multiplied by D on the right or on the left.

- ② Find a 3×3 matrix B , not the identity matrix or the zero matrix, such that $AB = BA$.

7.

Let $A = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix}$, $\mathbf{b}_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$, $\mathbf{b}_3 = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$, and $\mathbf{b}_4 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$. $A^{-1} = \begin{bmatrix} 6 & -1 \\ -\frac{5}{2} & \frac{1}{2} \end{bmatrix}$

- a. Find A^{-1} , and use it to solve the four equations $A\mathbf{x} = \mathbf{b}_1$, $A\mathbf{x} = \mathbf{b}_2$, $A\mathbf{x} = \mathbf{b}_3$, $A\mathbf{x} = \mathbf{b}_4$

- b. The four equations in part (a) can be solved by the same set of row operations, since the coefficient matrix is the same in each case. Solve the four equations in part (a) by row reducing the augmented matrix $[A \quad \mathbf{b}_1 \quad \mathbf{b}_2 \quad \mathbf{b}_3 \quad \mathbf{b}_4]$

$$A\mathbf{x} = \mathbf{b} \Rightarrow \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 2 & 3 \\ 3 & -5 & 6 & 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{bmatrix} = \begin{bmatrix} 6 & -1 \\ -\frac{5}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -1 & 1 & 2 & 3 \\ 3 & -5 & 6 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -9 & 11 & 6 & 13 \\ 4 & -5 & -2 & -5 \end{bmatrix} \#$$

8.

Suppose P is invertible and $A = PBP^{-1}$. Solve for B in terms of A .

$$\therefore B = P^{-1}AP$$

9. 10. Find the inverses of the matrices below individually, if they exist.

Use the algorithm introduced in this section.

$$\begin{bmatrix} 1 & -2 & 1 \\ 4 & -7 & 3 \\ -2 & 6 & -4 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$$

7. (b)

$$A[\mathbf{b}_1 \quad \mathbf{b}_2 \quad \mathbf{b}_3 \quad \mathbf{b}_4]$$

$$= \begin{bmatrix} 5 & -9 & 14 & 13 \\ 31 & -55 & 82 & 75 \end{bmatrix} \#$$

$$A\mathbf{b}_1 = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 31 \end{bmatrix}$$

$$A\mathbf{b}_2 = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ -5 \end{bmatrix} = \begin{bmatrix} -9 \\ -55 \end{bmatrix}$$

$$A\mathbf{b}_3 = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 14 \\ 82 \end{bmatrix}$$

$$A\mathbf{b}_4 = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 13 \\ 75 \end{bmatrix}$$

9. $[A \ I] \sim [I \ A^{-1}]$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 4 & -7 & 3 & 0 & 1 & 0 \\ -2 & 6 & -4 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \downarrow \times 2 \\ \downarrow \times (-4) \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -4 & 1 & 0 \\ 0 & 2 & -2 & 2 & 0 & 1 \end{array} \right] \downarrow \times (-2)$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -4 & 1 & 0 \\ 0 & 0 & 0 & 10 & -2 & 1 \end{array} \right]$$

$\therefore \left[\begin{array}{ccc} 1 & -2 & 1 \\ 4 & -7 & 3 \\ -2 & 6 & -4 \end{array} \right]^{-1}$ not exist

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \downarrow \times 3 \\ \downarrow \times (-2) \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{array} \right] \downarrow \times 3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array} \right] \times \frac{1}{2}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 1 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{array} \right] \begin{array}{l} \uparrow \times 2 \\ \uparrow \times 2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 1 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{array} \right]$$

$$\left[\begin{array}{ccc} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{array} \right]^{-1} = \left[\begin{array}{ccc} 8 & 3 & 1 \\ 10 & 4 & 1 \\ \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{array} \right] \#$$